ece4-exp

EC-Earth4 Experiment Configuration Made Simple

From hours of YAML editing to 30 seconds

The Problem

Creating an EC-Earth4 experiment config is painful:

- 📝 Edit 257-line YAML file manually
- 🧮 Calculate node layouts and processor distribution
- 🔧 Match components, grids, platform paths
- ⚠️ Easy to make mistakes (typos, incompatible combinations)
- ⏱️ Repeat for every experiment

Result: 2-4 hours for experienced users, longer for newcomers

The Solution

One command generates production-ready configs:

```
./ece4-exp generate \
  --recipe gcm-sr.yml \
  --sim-procs 1120 \
  --expid my-experiment
```

That's it. You get:

- ✅ Correct component configuration (OIFS + NEMO + XIOS + OASIS)
- ✅ Calculated node layout (10 nodes, 112 procs/node)
- ✅ Platform paths from upstream EC-Earth4 repo
- ✅ Validated and ready to run

Time saved: ~3.5 hours per experiment

Pre-Built Recipes

Ready-to-use experiment templates:

Recipe	Description	Procs	Nodes*
`gcm-sr.yml`	Coupled atmosphere-ocean	1120	10
`omip-sr.yml`	Ocean-only with ERA5 forcing	224	2
`amip-sr.yml`	Atmosphere-only (prescribed SST)	896	8
`ccycle-sr.yml`	Carbon cycle (with LPJG)	1120+	10+

*MareNostrum5 (112 cores/node)

All recipes: Tested, validated, production-ready

How It Works

YAML overlay system - configurations merged in order:

```
Base config (EC-Earth4 repo)
  ↓
+ Platform launcher (local file)
  ↓
+ Recipe (experiment pattern)
  ↓
+ Your CLI parameters
  ↓
= Generated config (229 lines)
```

Smart features:

- Auto-fetches latest base config from EC-Earth4 repo
- Auto-calculates processor distribution
- Validates before you submit

Real Examples

Coupled atmosphere-ocean (10 nodes):

```
./ece4-exp generate --recipe gcm-sr.yml --sim-procs 1120
# → OIFS + NEMO + XIOS + OASIS + RNFM
# → TL255L91 + eORCA1L75
```

Ocean-only (2 nodes):

```
./ece4-exp generate --recipe omip-sr.yml --sim-procs 224
# → NEMO + XIOS (no atmosphere)
# → Forced by ERA5
```

Different science, different resources - **same simple command**.

Setup: Once and Forever

One-time installation (2 minutes):

```
git clone https://github.com/vlap/ece4-exp.git
cd ece4-exp
./setup.sh
```

Configure your defaults (1 minute):

```
./ece4-exp init-user
# Edit ~/.config/ece4-exp/defaults.yml
```

Set your platform, account, scratch path **once**.

Then just:

```
./ece4-exp generate --recipe <type> --sim-procs <n>
```

Override when needed: `--platform ecmwf-hpc2020`

Advanced Features

Create custom recipes:

```
# Modify generated config
vim myexp.yml

# Save as new recipe
./ece4-exp save --expid myexp -o my-recipe.yml

# Reuse it
./ece4-exp generate --recipe my-recipe.yml --sim-procs 1120
```

Scripting support:

```
# Quiet mode (no colors, for scripts)
./ece4-exp generate --recipe gcm-sr.yml --sim-procs 1120 --quiet

# Batch generation
for exp in {001..010}; do
  ./ece4-exp generate --recipe gcm-sr.yml --expid exp$exp
```

Platform Support

Currently supported:

- 🇪🇸 BSC MareNostrum 5
- 🇮🇹 ECMWF HPC2020

Adding a new platform:

1. Create `platforms/<platform>.yml` with launcher templates
2. Specify `ppn` (processors per node)
3. Done!

Example:

```
ppn: 128  # Processors per node
slurm-wrapper-taskset:
  oifs:
    script: "srun --cpus-per-task=1 ..."
```

Get Started

Try it now:

```
git clone https://github.com/vlap/ece4-exp.git
cd ece4-exp
./QUICK_DEMO.sh  # Interactive 3-minute demo
```

Documentation:

- `README.md` - Complete guide
- `DEMO.md` - Walkthrough with examples
- `./ece4-exp --help` - Command reference

Compatible with:

- ✅ Manual EC-Earth4 workflows (ScriptEngine)
- ✅ Autosubmit (auto-ecearth4) - backward compatible